EFY TESTED
Ashwini K. Sinha

GPS TRACKER With Arduino

RAJESH RATHAKRISHNAN

A GPS (global positioning system) tracker utilises satellite signals to accurately determine the location of an object or person in real time. Originally developed by the United States Department of Defense for military applications, GPS technology has become widely accessible for civilian use. GPS trackers consist of hardware and software components that work together to receive signals from GPS satellites, calculate precise coordinates, and transmit this information to a central server or user device. They are commonly used in various industries such as transportation, logistics, fleet management, personal safety, and asset tracking. GPS trackers can be installed in vehicles, smartphones, wearable devices, or attached to assets to monitor their movement, location, and other relevant data. In recent years, the widespread availability of GPS tracking technology has led to its adoption in consumer-oriented applications as well.

Here, this GPS tracking device is designed to be installed in robots and vehicles and programmed according to the needs of people. The author's prototype is shown in Fig. 1. The components required to build this tracker are listed in Bill of Materials table. Main components used in this tracker are shown in Fig. 2.

BILL OF MATERIALS

Components	Quantity
Arduino Uno (MOD1)	1
GPS module NEO 6M or similar GPS (MOD2)	1
Jumper wires	10
USB C adaptor	1

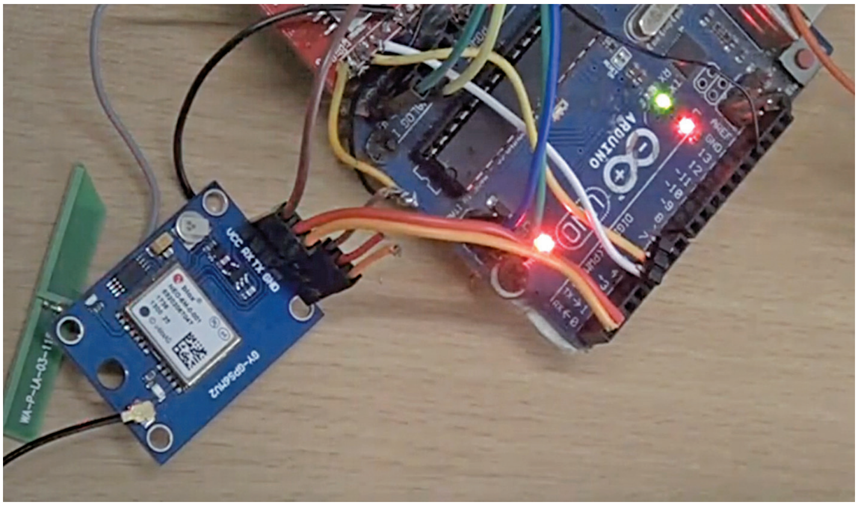

Fig. 1: Author's prototype

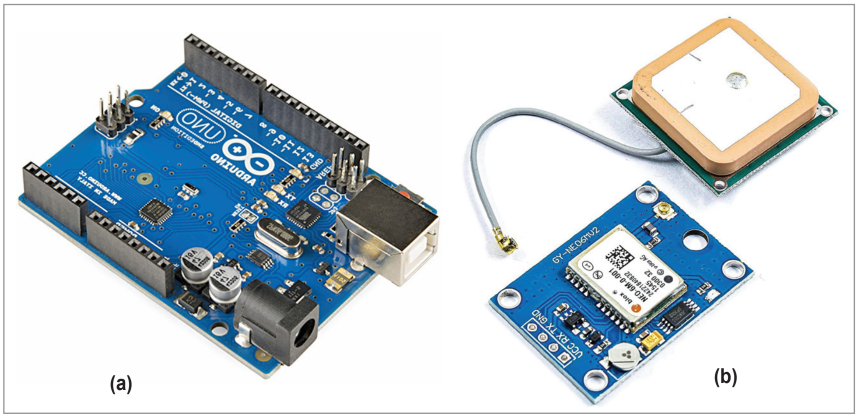

Fig. 2: (a) Arduino board, and (b) NEO-6M GPS module

Arduino board. The Arduino Uno is one of the most popular microcontroller boards in the Arduino line-up. It is based on the ATmega328P microcontroller and provides a simple and accessible platform for electronics enthusiasts, hobbyists, and professionals to cre-

Receiver Type: 50-channel u-blox 6 positioning engine
Satellite Systems: GPS, GLONASS, Galileo, BeiDou
Position Accuracy:
Horizontal: <2.5 metres (GPS/GNSS positioning)
Vertical: <4.0 metres (GPS/GNSS positioning)
Update Rate: Up to 5Hz (configurable)
Sensitivity:
Tracking: -161dBm
Reacquisition: -160dBm
Cold Start: -147dBm
Operating Voltage: 3.0V to 5.0V DC
Interfaces: UART (TTL-level), default baud rate: 9600bps
Antenna Interface: SMA or u.fl connector
Operating Temperature: -40°C to +85°C

Fig. 3: NEO-6M GPS module specifications